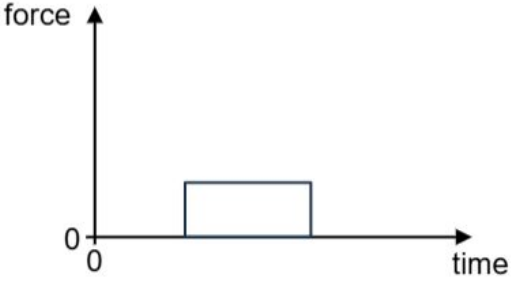
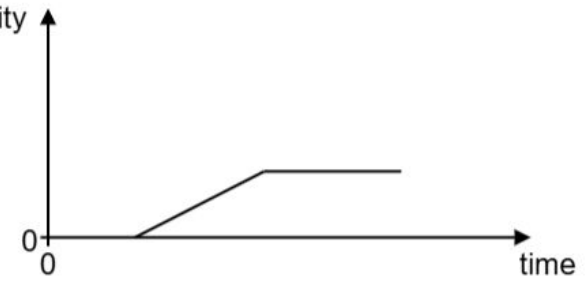
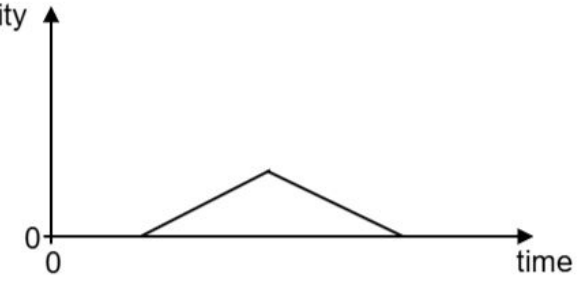
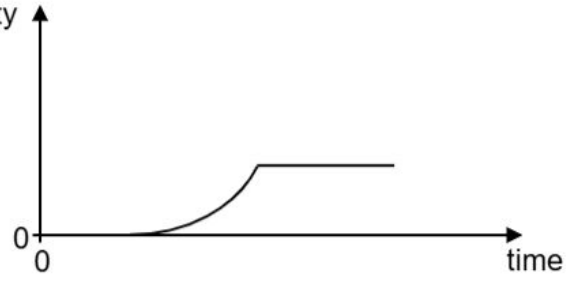
5	A body is initially at rest on a horizontal frictionless table. A force then acts on it that varies with time as shown.  Which graph represents the variation with time of the velocity of the body?

	A	
	B	
	C	
	D	
L2	**Answer: B** According to Newton's Second Law of Motion, F_{net} = ma, (i) When net force is constant, the acceleration is constant. The rate of change of velocity will be constant and hence the v-t graph is a straight line with non-zero gradient. (ii) When net force is zero, the acceleration is zero. The rate of change of velocity will be zero and hence the v-t graph is a straight line with zero gradient.	